

```

    }

}

i += EVENT_SIZE + event->len;
}

}

/* Clean up*/
inotify_rm_watch( fd, wd );
close( fd );

return 0;
}

```

Given below is the output of compiling and running Listing 3:

```

$ gcc auto_compile.c
$ ./a.out /home/gene/codes/
Watching:: /home/gene/codes/
Compiling:: hello1.c
Compiling:: hello2.c

```

Of course, you could easily modify the above code to compile/execute programs in other languages.

Notify tools

Inotify-tools (<https://github.com/rvoicilas/inotify-tools/wiki>) is a C library, as well as a set of command-line programs providing a user-friendly interface for Inotify. One useful tool is *inotifywatch*, which can be used to watch a directory, and is thus a more feature-rich version of our last exercise (Listing 2).

dnotify, inotify and fanotify

Inotify succeeded *dnotify*, and *fanotify*

is slowly gathering momentum, having been introduced in kernel 2.6.36. See <http://lwn.net/Articles/360955/> for a description of it.

If you navigate to the kernel sources sub-directory under *fs/notify*, you can see implementations of the above three co-existing in a possible state of harmony.

In this article, we have used a hands-on approach to study a very useful feature of the Linux kernel, and all in user-space. Though not a particularly new concept (in use since 2005), now you know how you can write applications that perform actions in response to filesystem events. If you program in a language other than C, there are APIs available for C++, Java, Python, Ruby, and even a *node.js* module! Here are a few suggestions for building upon the code given in this article:

- Extend *auto_compile.c* to be implemented as a file-manager extension, such as a *Nautilus* extension, so that whenever a new source file is created or modified in a designated directory, it automatically compiles it.
- Code using Inotify will be most useful when implemented as a *daemon* process, such that it can run in the background without any user intervention required. Hence, the *while(1)* loop can be better implemented as a 'wake-on' feature, such that at other times, it stays in a sleeping mode. 

Resources

- [1] <http://www.ibm.com/developerworks/linux/library/l-ubuntu-inotify/index.html>
- [2] Robert Love's article: <http://www.linuxjournal.com/article/8478?page=0,0>
- [3] Manual page: <http://linux.die.net/man/7/inotify>
- [4] Code listings for this article are available from: https://bitbucket.org/amitsaha/articles_code/src

By: Amit Saha

The author blogs at <http://amitsaha.wordpress.com> and welcomes queries and suggestions at amitsaha.in@gmail.com.

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